Oncology-informed Super-colocation Detection for Hypothesis Generation in Immune Checkpoint Inhibitor Therapy [Appendix]

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A Appendix

A.1 TME Map Generation

Figure 1 depicts how TME maps are generated using MxIF imaging [1, 2]: 1) Specimen handling. The tissue biopsy is stabilized by chemicals, then dehydrated and embedded in wax. A thin slice of approximately 5 µm in depth is taken and placed on glass slides. 2) Multiplex staining. Each slide is stained with multiple antibodies. Each antibody is paired with a unique fluorescent dye and recognizes a specific biomarker, such as a CD8+ protein. 3) Image acquisition. After staining, slides are scanned with multispectral imaging systems. 4) Image processing. This step applies flat-field and lens-distortion corrections to improve image quality. 5) Phenotyping. With the processed image, each cell is masked, and its coordinates are logged. Pixel intensities in each channel under the mask determine whether the cell expresses the channel-specific biomarker. Cellular biomarkers determine cell types. 6) Normalizing batch effects. This step uses control tissue in each batch to normalize pixel intensities slide by slide. 7) Data handling. The final single-cell-resolution dataset contains the coordinates and expressed markers for each cell.

A.2 Three-Feature Hypothesis Generation Results

Table 1 lists the most promising three-feature oncology-informed super-colocations that can serve as hypotheses for further lab investigations. Note that we only show patterns that have relatively high differences in the interaction intensities (measured by JSD>=0.1) between responders and non-responders.

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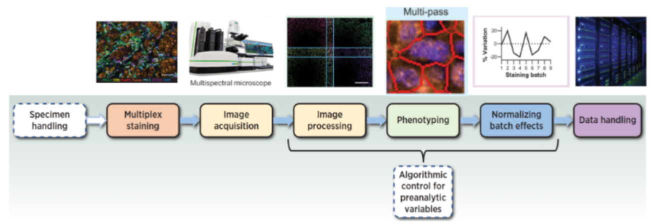

Figure 1: The process to generate TME maps [1, 2].

Table 1: Three-Feature Super-Colocations/Hypotheses.

Cell Combinations	JS Distance
Tumor, Other, CTL	0.2260
Tumor, Other, B cell	0.1877
Tumor, Other, Dendritic cell	0.1814
Tumor, Other, Helper T-cell	0.1802
Tumor, Other, Classical Monocyte	0.1740
Tumor, Other, Nonclassical Monocyte	0.1675
Tumor, Other, M2 Macrophage	0.1637
Tumor, Other, Regulatory T-cell	0.1631
Tumor, Other, Lymph and Blood Vessel	0.1628
Tumor, Other, Neutrophil	0.1584
Tumor, Other, M1 Macrophage	0.1568
Tumor, CTL, Helper T-cell	0.1555
Tumor, Other, NK Cell	0.1554
Tumor, Classical Monocyte, CTL	0.1539
Tumor, B cell, CTL	0.1490
Tumor, CTL, Dendritic cell	0.1401
Tumor, M2 Macrophage, CTL	0.1353
Tumor, CTL, Lymph and Blood Vessel	0.1281
Tumor, Nonclassical Monocyte, CTL	0.1273
Tumor, CTL, Regulatory T-cell	0.1257
Tumor, M1 Macrophage, CTL	0.1232
Tumor, CTL, Neutrophil	0.1231
Tumor, CTL, NK Cell	0.1222
Tumor, Classical Monocyte, Dendritic cell	0.1153
Tumor, Classical Monocyte, B cell	0.1135
Tumor, B cell, Dendritic cell	0.1050

A.3 Sensitivity Analysis on Anchor Radii

Table 2: Ranking of Two-feature OISCs across Different Anchor Radii (μm).

Ranking	**15** μm	**20** μm	**25** μm	**50** μm	**100** μm
1	Other	Other	Other	CTL	CTL
2	CTL	CTL	CTL	Other	Other
3	Classical Monocyte	Classical Monocyte	Classical Monocyte	Classical Monocyte	Classical Monocyte
4	B cell	Dendritic cell	Dendritic cell	Dendritic cell	Lymph and Blood Vessel
5	Dendritic cell	B cell	B cell	B cell	Dendritic cell
6	Lymph and Blood Vessel	Helper T-cell	Helper T-cell	Lymph and Blood Vessel	B cell
7	Helper T-cell	Lymph and Blood Vessel	Lymph and Blood Vessel	Helper T-cell	M1 Macrophage
8	Nonclassical Monocyte	M2 Macrophage	Nonclassical Monocyte	Nonclassical Monocyte	Helper T-cell
9	M2 Macrophage	Nonclassical Monocyte	M2 Macrophage	M1 Macrophage	Regulatory T-cell
10	Regulatory T-cell	Regulatory T-cell	M1 Macrophage	M2 Macrophage	Nonclassical Monocyte
11	Neutrophil	M1 Macrophage	Regulatory T-cell	Regulatory T-cell	M2 Macrophage
12	M1 Macrophage	Neutrophil	Neutrophil	Neutrophil	Neutrophil
13	NK cell	NK cell	NK cell	NK cell	NK cell

Table 3: Ranking of Three-feature OISCs across Different Anchor Radii (μm)

Ranking	**15** μm	**20** μm	**25** μm	**50** μm	**100** μm
1	Other	Other	Other	Other	Other
2	Classical Monocyte	Helper T-cell	Helper T-cell	Helper T-cell	B cell
3	Helper T-cell	Classical Monocyte	Classical Monocyte	B cell	Helper T-cell
4	B cell	B cell	B cell	Classical Monocyte	M2 Macrophage
5	Dendritic cell	Dendritic cell	Dendritic cell	M2 Macrophage	Lymph and Blood Vessel
6	M2 Macrophage	M2 Macrophage	M2 Macrophage	Dendritic cell	Classical Monocyte
7	Lymph and Blood Vessel	Lymph and Blood Vessel	Nonclassical Monocyte	Lymph and Blood Vessel	Neutrophil
8	Nonclassical Monocyte	Nonclassical Monocyte	Lymph and Blood Vessel	Regulatory T-cell	Regulatory T-cell
9	Regulatory T-cell	Regulatory T-cell	Regulatory T-cell	Nonclassical Monocyte	Dendritic cell
10	Neutrophil	M1 Macrophage	M1 Macrophage	Neutrophil	M1 Macrophage
11	M1 Macrophage	Neutrophil	Neutrophil	M1 Macrophage	Nonclassical Monocyte
12	NK cell	NK cell	NK cell	NK cell	NK cell

References

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